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Studierichting: Informatica

Oefeningenreeks 1C nr. 45.3

$$z^3 = -i = 0 - 1i$$

$$\text{modulus: } r = \sqrt{0^2 + (-1)^2} = 1$$

$$\text{argument: } \tan \theta = \frac{-1}{0} \Rightarrow \theta = \frac{\pi}{2}$$

$$z^3 = \text{cis} \left(-\frac{\pi}{2} + k \cdot 2\pi \right) \Rightarrow z = \text{cis} \left(\frac{-\frac{\pi}{2} + k \cdot 2\pi}{3} \right) \quad \forall k \in \{0, 1, 2\}$$

$$k = 0 \rightarrow z = \text{cis} \left(\frac{-\frac{\pi}{2} + 0 \cdot 2\pi}{3} \right) = \text{cis} \left(-\frac{\pi}{6} \right) = \frac{\sqrt{3}}{2} - \frac{1}{2}i$$

$$k = 1 \rightarrow z = \text{cis} \left(\frac{-\frac{\pi}{2} + 1 \cdot 2\pi}{3} \right) = \text{cis} \left(\frac{\frac{3\pi}{2}}{3} \right) = \text{cis} \left(\frac{\pi}{2} \right) = i$$

$$k = 2 \rightarrow z = \text{cis} \left(\frac{-\frac{\pi}{2} + 2 \cdot 2\pi}{3} \right) = \text{cis} \left(\frac{\frac{7\pi}{2}}{3} \right) = \text{cis} \left(\frac{7\pi}{6} \right) = -\frac{\sqrt{3}}{2} - \frac{1}{2}i$$

References